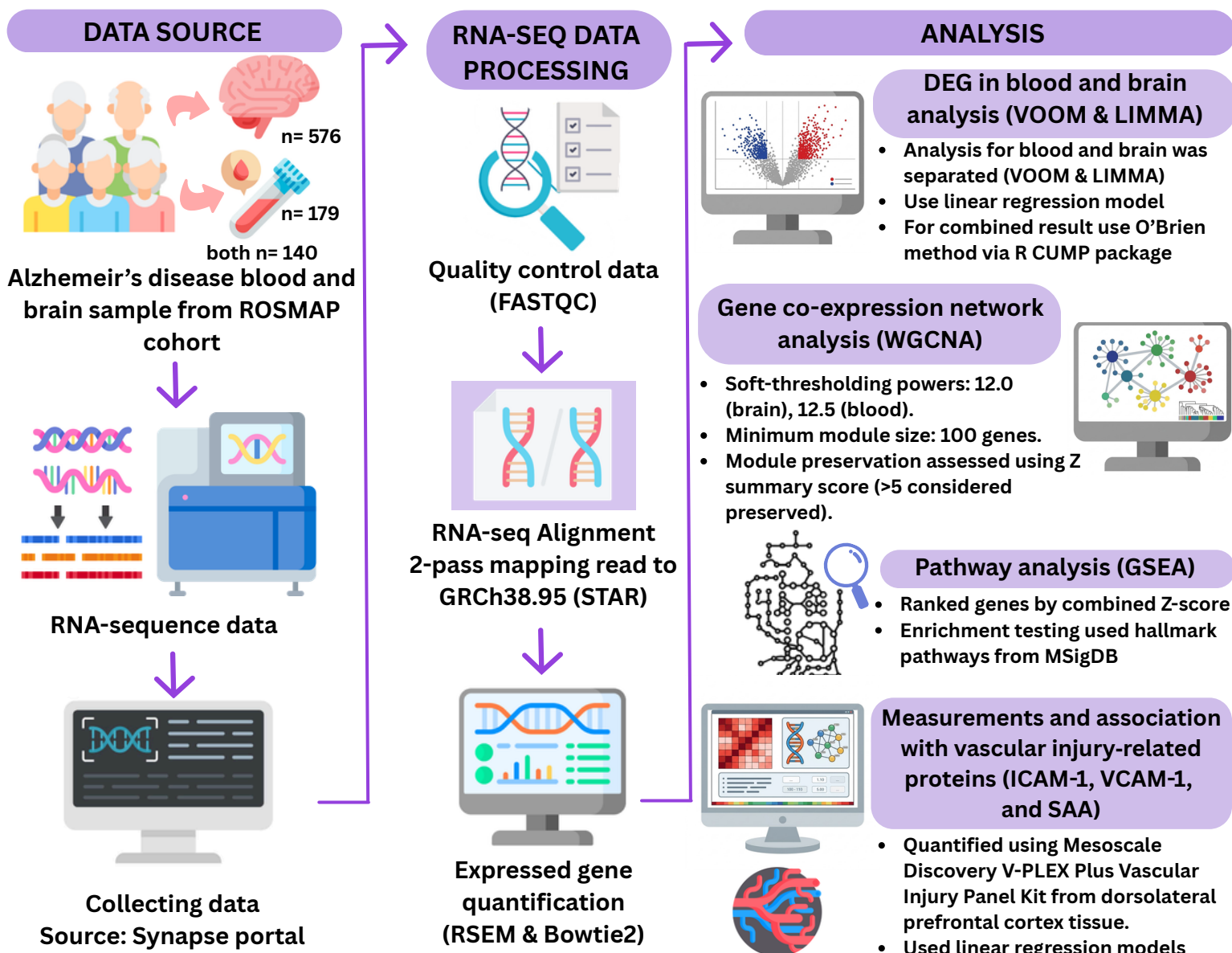




“BLOOD AND BRAIN TRANSCRIPTOME ANALYSIS REVEALS APOE GENOTYPE-MEDIATED AND IMMUNE-RELATED PATHWAYS INVOLVED IN ALZHEIMER DISEASE” INFOGRAPHIC

METHOD



KEY FINDINGS

Differentially Expressed Genes (DEGs)

- HLA-DQA1 and INPP5D of 64 AD-related genes (from GWAS) expressed in **both** blood and brain significantly ($P < 0.05$) and showed consistent direction of effect: **HLA-DQA1** (downregulated) & **NPP5D** (upregulated).
- HLA-DQA1 and INPP5D highly expressed in blood dendritic cells and monocytes.
- INPP5D specifically expressed in brain microglia.

APOE Genotype-Dependent Pathways

- 21 pathways** significantly enriched in at least one APOE genotype group.
- 6 pathways unique to $\epsilon 3/\epsilon 4$ carriers: Apoptosis, estrogen response late, hypoxia, IL-6/JAK/STAT3 signaling, inflammatory response, & p53 pathway.
- 4 pathways with opposite enrichment direction between $\epsilon 4$ carriers and non-carriers: Allograft rejection, interferon gamma response, peroxisome, & TNFA signaling via NFKB.

Co-Expression Networks

- Found “**Light Green**” network preserved in Blood and Brain
- Significantly enriched for AD risk genes and plaque score-associated genes.
- Eigengene value higher in APOE $\epsilon 4$ carriers vs. non-carriers ($P = 4.7 \times 10^{-3}$).
- Enriched in 9 KEGG pathways and 4 hallmark pathways.
- Key genes: **NFKBIA**, **HLA-DRA**, **INPP5D**, **C4B**, **VASP**.

Association with vascular injury proteins

- From Light Green network: **VASP** was associated with ICAM-1 and **C4B** was associated with ICAM-1, SAA, & VCAM-1.
- from TNFA/ NF- κ B pathway: **TRIP10** & **FOSL1** was associated with ICAM-1.

Source:

Panitch, R., Hu, J., Xia, W. et al. 2022. Blood and Brain Transcriptome Analysis Reveals APOE Genotype-mediated and Immune-related Pathways Involved in Alzheimer Disease. *Alzheimer's Research & Therapy*. 14 (30). <https://doi.org/10.1186/s13195-022-00975-z>